

The bicaption package*

Axel Sommerfeldt

<https://gitlab.com/axelsommerfeldt/caption>

2021/01/04

Abstract

This package supports the typesetting of bilingual captions.

Contents

1	Loading the package	2
2	Setting options	2
3	Additional options	3
4	The <code>\bicaption</code> commands	4
5	Customising lists	5
6	Language Selection	7
7	Supported packages	10
7.1	Support of the listings package	10
7.2	Support of the longtable package	11
7.3	Support of the subcaption package	11

*This package has version number v1.5.

1 Loading the package

This package will be loaded by

```
\usepackage[\langle options \rangle]{bication} .
```

The options for the bication package are the same ones as for the caption package and specify settings which are used for the second language *additionally*. In fact

```
\usepackage[\langle options \rangle]{bication}
```

is identical to

```
\usepackage{bication}  
\captionsetup[bi-second]{\langle options \rangle} .
```

When used with the babel or polyglossia package, the bication package should be loaded *after* it, so the main language will be set automatically. See [section 6](#) for details.

2 Setting options

`\captionsetup` The command

```
\captionsetup[bi]{\langle options \rangle}
```

does setup options which will be used for bilanguage captions *additionally* to the ones which are setup for the specific floating environment.

```
\captionsetup[bi-first]{\langle options \rangle}
```

does setup options which will be used for the *first* heading of the bilanguage captions *additionally* to the ones which are setup for the specific floating environment and the ones which are setup by `\captionsetup[bi]{...}`.

```
\captionsetup[bi-second]{\langle options \rangle}
```

does setup options which will be used for the *second* heading of the bilanguage captions *additionally* to the ones which are setup for the specific floating environment and the ones which are setup by `\captionsetup[bi]{...}`.

Options specified with `\usepackage[...]{bication}` and `\captionsetup[bi...]{...}` will override the ones specified by `\captionsetup{...}` and `\captionsetup[figure]{...}` (same for ‘table’). So finally we have the following order how settings for bilingual captions are applied:

1. Global settings (`\usepackage[...]{caption}` and `\captionsetup{...}`)
2. Environmental settings (`\captionsetup[figure -or- table]{...}`)
3. Local settings (`\captionsetup{...}` inside figure or table environment)

4. Custom ‘bi’ settings (`\captionsetup[bi]{...}`)
5. Custom ‘bi-first’ resp. ‘bi-second’ settings (`\usepackage[...]{bication}` and `\captionsetup[bi-first]{...}` resp. `\captionsetup[bi-second]{...}`)

An example:

```
\usepackage[labelsep=quad,indentation=10pt]{caption}
\usepackage[labelfont=bf]{bication}
\captionsetup[table]{labelfont=it,position=top}
```

causes the second heading of the bilingual caption inside `table` environments to be typeset with the settings

```
labelsep=quad,indentation=10pt,position=top,labelfont=bf.
```

To limit `bi`, `bi-first`, or `bi-second` options to specific environments one can use multiple optional arguments for `\captionsetup`, e.g.:

```
\captionsetup[figure][bi-first][<options>]
```

will limit the settings to the first heading of `figure` environments only. Please note that the environment name (`figure`, `table`, ...) has to be specified as first optional argument while the bilingual selection (`bi`, `bi-first`, or `bi-second`) as second one.

3 Additional options

These options are available additional to the ones offered by the `caption` package:

- | | |
|-----------------------|--|
| <code>lang=</code> | <p>Sets the language of the caption, e.g.</p> <pre>\usepackage[lang=english]{bication}</pre> <p>will typeset the second caption of bilingual captions in English. (The language will be set with <code>\selectcaptionlanguage</code> internally, see section 6 for details.)</p> |
| <code>bi-lang=</code> | <p>Causes a selection of the headings of bilingual captions.</p> <pre>\captionsetup{bi-lang=both}</pre> <p>will cause that both caption headings are being typeset. (This is the default.)</p> <pre>\captionsetup{bi-lang=first}</pre> <p>will cause that only the <i>first</i> heading is being typeset, and</p> <pre>\captionsetup{bi-lang=second}</pre> <p>will cause that only the <i>second</i> heading is being typeset.</p> |

$$\text{bi-slc}=\qquad\qquad\backslash\text{captionsetup}\{\text{bi-slc}=\langle\textit{bool}\rangle\}$$

switches the common single-line-check `on` or `off`, i.e. when switched on only a single check will be done for both captions, and the result will affect both captions afterwards. So if only one caption is longer than a single line, both captions will be treated as if they are longer than a single line, even if the second one isn't. (The default is `on`.)

$$\text{bi-swap} = \text{\captionsetup{bi-swap}}$$

will swap the primary and secondary language, making the first language the second one and vice versa. (The default is `false`.)

New feature
v1.3

$$\text{bi-separator}=\quad \backslash\text{captionsetup}\{\text{bi-separator}=\langle name\rangle\}$$

will select a separator between first and second bilingual caption. You could choose one of the following: ‘none’ (which is the default one and could also be addressed as ‘default’), ‘smallskip’, ‘medskip’, ‘largeskip’, or a self-defined one using

DeclareBiCaptionSeparator

$$\backslash\text{DeclareBiCaptionSeparator}\{\langle name \rangle\}\{\langle code \rangle\}$$

Examples:

\captionsetup{bi-separator=smallskip}

will put a `\smallskip` between the two bilingual captions.

```
\DeclareBiCaptionSeparator{hrule}{\hrule}
\captionsetup{bi-separator=hrule}
```

will draw a horizontal line between the two bilingual captions.

```
\DeclareBiCaptionSeparator{3pt}{\vspace{3pt}}
\captionsetup{bi-separator=3pt}
```

will put 3pt vertical space between the two bilingual captions.

4 The `\bicaption` commands

`\bicaption` Bilingual captions will be typeset by

```
\bicaption[⟨list entry #1⟩]{⟨heading #1⟩}
           [⟨list entry #2⟩]{⟨heading #2⟩}
\bicaption*{⟨heading #1⟩}{⟨heading #2⟩}
```

The `\label` should be placed either after this command, or inside the first heading.

`\bicaptionbox` Bilingual caption boxes will be typeset by

```

\bicaptionbox[⟨list entry #1⟩]{⟨heading #1⟩
[⟨list entry #2⟩]{⟨heading #2⟩
[⟨width⟩][⟨inner-pos⟩]{⟨contents⟩}
\bicaptionbox*{⟨heading #1⟩}{⟨heading #2⟩
[⟨width⟩][⟨inner-pos⟩]{⟨contents⟩}

```

The `\label` should be placed inside the first heading.

(For a description of the optional parameters *<width>* and *<inner-pos>* please take a look at the caption package documentation, `\captionbox`.)

5 Customising lists

`list=` As default both caption texts will be insert into the List of Figures resp. List of Tables. To suppress the second entry just pass the option `list=off` to the `bicaption` package, e.g.:

```
\usepackage[lang=english,...,list=off]{bicaption}
```

`listtype+=` Another option is separating the lists. For that purpose the option

```
listtype+=<list type extension>
```

can be used to tell the `bicaption` package to use a different list for the second caption text. The given value will be appended to the current environment type; for example with `listtype+=X` the list entries will be put into the list responsible for the types `figureX` ($= \text{figure} + X$), `tableX` ($= \text{table} + X$) etc.

Such a *<list type>* can be defined using `\DeclareFloatingEnvironment` offered by the `newfloat` package, but some document classes or other packages offer macros for defining new floating environment types (and their corresponding lists) as well.

A sample document:

```
\documentclass[a4paper]{article}

% Use "ngerman" as 1st language, "english" as 2nd one
\usepackage[english,ngerman]{babel}

% Load the bicaption package with 2nd language set to
% "english", and list type "figureEng" resp. "tableEng"
\usepackage[lang=english,listtype+=Eng]{bicaption}

\usepackage{newfloat}
% Define type "figureEng" and \listoffigureEng
\DeclareFloatingEnvironment[fileext=lof2]{figureEng}
    [Figure][List of Figures]
% Define type "tableEng" and \listoftableEng
\DeclareFloatingEnvironment[fileext=lot2]{tableEng}
    [Table][List of Tables]

\begin{document}
\listoffigures      % typeset "Abbildungsverzeichnis"
\listoffigureEng    % typeset "List of Figures"

\begin{figure}
\centering
A placeholder for an image or whatever
\bicaption{Deutscher Text}{English text}
```

```

\end{figure}

\end{document}

```

A different approach is using one list for both languages, but with different formatting. Since the `caption` package does not offer options and commands for customising the format of the lists, one needs an additional package for this purpose, for example the `titletoc` package:

```

\documentclass[a4paper]{article}

% Use "ngerman" as 1st language, "english" as 2nd one
\usepackage[english,ngerman]{babel}

% Load the bicaption package with 2nd language set to
% "english", and list type "figure2" resp. "table2"
\usepackage[lang=english,listtype+=2]{bicaption}

% We load the titletoc package for customizing lists
% Note: Loading titletoc should be done prior
% defining additional floating environments with
% \DeclareFloatingEnvironment
\usepackage{titletoc}

\usepackage{newfloat}
% Define the new floating environment type "figure2"
% Use the same file extension as for "figure" (.lof) here
\DeclareFloatingEnvironment[fileext=lof]{figure2}
% Define the new floating environment type "table2"
% Use the same file extension as for "table" (.lot) here
\DeclareFloatingEnvironment[fileext=lot]{table2}

% We use the titletoc package for customizing "figure2"
% which is appropriate for the second language captions
\titlecontents{figure2}[3.8em]
{} % no above code
{} % empty numbered entry format
{} % empty numberless entry format
{} % empty filler page format

\begin{document}
\renewcommand\listfigurename
{Abbildungsverzeichnis / List of Figures}
\listoffigures

\begin{figure}
\centering
A placeholder for an image or whatever
\bicaption{Deutscher Text}{English text}
\end{figure}

```

```
\end{document}
```

6 Language Selection

For language selection the bcaption package uses two macros internally:

`\captionmainlanguage` `\captionmainlanguage` contains the main language, e.g. `english` or `german`. If not set prior to loading the bcaption package, the bcaption package will try to obtain this setting from the babel or polyglossia package.

So if you are using either babel or polyglossia, and want to adopt the main language setting from it, then just load the bcaption package *after* it, and simply forget about the `\captionmainlanguage` stuff.

Otherwise one can either define `\captionmainlanguage` prior to loading the bcaption package, e.g.:

```
\newcommand\captionmainlanguage{french}
\usepackage[<options>]{bcaption}
```

Or one can specify the main language via `\captionsetup` after loading the bcaption package, e.g.:

```
\usepackage[<options>]{bcaption}
\captionsetup[bi-first]{lang=french}
```

When not using the babel or polyglossia package both approaches will have exactly the same effect. But when using the babel or polyglossia package, and one want to specify the main caption language manually, the first approach is preferable since defining `\captionmainlanguage` will suppress the automatic detection mechanism.

`\selectcaptionlanguage` `\selectcaptionlanguage` will be used internally to select the language:

New feature
v1.1

```
\selectcaptionlanguage{<font-or-list-entry>}{<language>}
```

For setting the language of the caption *<font-or-list-entry>* will be `\@firstoftwo`, for setting the language of the list entry *<font-or-list-entry>* will be `\@secondoftwo`.¹ It defaults to `\select@language (caption)` resp. `\selectlanguage (list entry)` offered by the babel and polyglossia package:

```
\providecommand*\selectcaptionlanguage[2]{%
  #1{\select@language}{\selectlanguage}{#2}}
```

If you need to alter this, just either define `\selectcaptionlanguage` prior loading the bcaption package, or redefine it afterwards.

¹`\@firstoftwo` and `\@secondoftwo` are defined in the L^AT_EX kernel and simply pick either the 1st or 2nd argument.

Example document using babel:

```
\documentclass[a4paper]{article}

% Use "ngerman" as 1st language, "english" as 2nd one
\usepackage[english,ngerman]{babel}

% Add custom translations to babel
\addto\captionsgerman{%
  % \renewcommand\whateveryname{Wasauchimmer}%
  % ...
}
\addto\captionsenglish{%
  % \renewcommand\whateveryname{Whatever}%
  % ...
}

% Load the bicaption package with 2nd language set to
% "english"
\usepackage[lang=english]{bicaption}

\begin{document}

\begin{figure}
  \centering
  A placeholder for an image or whatever
  \bicaption{Deutscher Text}{English text}
\end{figure}

\end{document}
```

The same example document but using a custom implementation of `\captionmainlanguage` and `\selectcaptionlanguage` instead of babel:

```
\documentclass[a4paper]{article}

% Set "german" as main bi-caption language
\newcommand\captionmainlanguage{german}

% Load the bicaption package with 2nd language set to
% "english"
\usepackage[lang=english]{bicaption}

% Declare an own language switching mechanism
% for bi-captions (instead of using babel)
\renewcommand\selectcaptionlanguage[2]{%
  \csname captions#2\endcsname}
\newcommand\captionsgerman{%
  \renewcommand\figurename{Abbildung}%
  \renewcommand\tablename{Tabelle}%
}
```

```

% ...
}
\newcommand\captionseenglish{%
  \renewcommand\figurename{Figure}%
  \renewcommand\tablename{Table}%
  % ...
}

\begin{document}

\begin{figure}
  \centering
  A placeholder for an image or whatever
  \bicaption{Deutscher Text}{English text}
\end{figure}

\end{document}

```

`\DeclareCaptionLangOption` For internal implementation reasons the selection of language will be done delayed, i.e. not done immediately at `lang=<language>`. So if you do

New feature
v1.2

```
\captionsetup[bi-second]{lang=ngerman,labelsep=quad}
```

the language `ngerman` will only be stored internally, and the label separator will be set to `quad` afterwards. Some time later, right before the caption is actually typeset, the language will be set to `ngerman`.

Usually this is no problem, but think of options which will be overwritten by the language selection, or options which act on the language currently set, for example

```
\captionsetup[bi-second]{lang=ngerman,name=Bild} .
```

`lang=ngerman` changes the environment name to “Abbildung”, and `name=Bild` changes the environment name to “Bild”. One would expect that the name is finally “Bild”, but because of the delayed nature of `lang=ngerman` it will be “Abbildung” instead, at least if we don’t take action about this.

For that reason the command

```
\DeclareCaptionLangOption{<caption option name>}
```

is offered. Options handled this way will be applied twice if used after the `lang=` option, when the option is actually used, and right after the language is selected.

```
\DeclareCaptionLangOption{name}
```

will be done by the `bicaption` package automatically, since the environment name will usually be overwritten by a language selection. So actually

```
\captionsetup[bi-second]{lang=ngerman,name=Bild}
```

will give the expected result, i.e. the environment name is typeset as “Bild”.

7 Supported packages

The caption package was adapted to the following packages which deals with captions, too: listings, longtable, and subcaption.

7.1 Support of the listings package

New feature
v1.5

If the listings package is loaded, the listings options `caption1` and `caption2` are available additionally, where option `caption1` specifies the caption of the first language and `caption2` of the second one.

Example document, using distinctive lists for each language:

```
\documentclass[a4paper]{article}
\usepackage{graphicx}

% Use "ngerman" as 1st language, "english" as 2nd one
\usepackage[english,ngerman]{babel}

% Load the bicaption package with 2nd language set to
% "english", and list type "figureEng" resp. "tableEng"
\usepackage[lang=english,listtype+=Eng,font=it]{bicaption}
\captionsetup{slc=off} % do not center short captions

\usepackage{listings}

% Change English name of "Listing" to "Code"
\addto\captionsenglish{\renewcommand\lstlistingname{Code}}

\usepackage{newfloat}
% Define the new floating environment type "lstlistingEng"
% (just to get an extra list for English listing captions)
\DeclareFloatingEnvironment[fileext=lol2]{lstlistingEng}
                                [Code][List of Codes]

\begin{document}

\lstlistoflistings           % German
\listoflstlistingEng        % English

\clearpage

\begin{lstlisting}
    [language=C,
    caption1=Deutscher Titel,
    caption2=English Title]
int main()
{
    printf( "Hello world!\n" );
    return 0;
}
```

```

    }
\end{lstlisting}

\end{document}

```

7.2 Support of the longtable package

If the `longtable` package is loaded, `\bicaption` is available in the `longtable` environment as well, e.g.:

```

\documentclass[a4paper]{article}

% Use "ngerman" as 1st language, "english" as 2nd one
\usepackage[english,ngerman]{babel}

% Load the bicaption package with 2nd language set to
% "english"
\usepackage[lang=english]{bicaption}

\usepackage{longtable}

\begin{document}

\begin{longtable}{ll}
  \bicaption{Deutscher Titel}{English Title}\\
  A & B \\
  C & D \\
  ...
\end{longtable}

\end{document}

```

7.3 Support of the subcaption package

If the `subcaption` package is loaded, these commands are available additionally:

`\bisubcaption` Bilingual sub-captions will be typeset by

```

\bisubcaption[⟨list entry #1⟩]{⟨heading #1⟩}
              [⟨list entry #2⟩]{⟨heading #2⟩}
\bisubcaption*{⟨heading #1⟩}{⟨heading #2⟩}

```

The `\label` should be placed either after this command, or inside the first heading.

`\bisubcaptionbox` Bilingual sub-caption boxes will be typeset by

```

\bisubcaptionbox[⟨list entry #1⟩]{⟨heading #1⟩}
                [⟨list entry #2⟩]{⟨heading #2⟩}
                [⟨width⟩][⟨inner-pos⟩]{⟨contents⟩}
\bisubcaptionbox*{⟨heading #1⟩}{⟨heading #2⟩}
                [⟨width⟩][⟨inner-pos⟩]{⟨contents⟩}

```

The `\label` should be placed inside the first heading.

(For a description of the optional parameters `<width>` and `<inner-pos>` please take a look at the `subcaption` package documentation, `\subcaptionbox`.)

7.3.1 A sample document

```
\documentclass[english,ngerman]{article}
\usepackage{selinput}
\SelectInputMappings{adieresis={ä},germandbls={ß}}

\usepackage{babel}
\usepackage[lang=english,font=it]{bicaption}
\usepackage[format=hang]{subcaption}

\begin{document}

\begin{figure}[!htb]
  \centering
  \bisubcaptionbox
    {Teilabbildung A\label{fig:test:A}}
    {Subfigure A}[0.4\textwidth]{IMAGE}%
  \quad
  \bisubcaptionbox
    {Teilabbildung langer Titel B\label{fig:test:B}}
    {Subfigure long title B}[0.4\textwidth]{IMAGE}%
  \bicaption{Deutscher Titel}{English Title}
  \label{fig:test}
\end{figure}

\captionsetup{bi-lang=both}

\begin{figure}[!htb]
  \centering
  \bisubcaptionbox[A]
    {Und eine gaaaanz lange Caption: Teilabbildung A}
    {Subfigure A}[0.4\textwidth]{IMAGE}%
  \quad
  \bisubcaptionbox[B]
    {Teilabbildung B}
    {Subfigure B}[0.4\textwidth]{IMAGE}%
  \bicaption[Abbildungsverzeichnistitel]
    {Und eine noch viel viel viel
     längere deutsche Beschriftung: Deutscher Titel}
    {Short English heading}
\end{figure}

\captionsetup{bi-slc=off}

\begin{figure}[!htb]
```

```

\centering
\bisubcaptionbox[A]
  {Und eine gaaaanz lange Caption: Teilabbildung A}
  {Subfigure A}[0.4\textwidth]{IMAGE}%
\qquad
\bisubcaptionbox[B]
  {Teilabbildung B}
  {Subfigure B}[0.4\textwidth]{IMAGE}%
\bicaption[Abbildungsverzeichnistitel]
  {Und eine noch viel viel viel
   längere deutsche Beschriftung: Deutscher Titel}
  {Short English heading}
\end{figure}

\captionsetup{slc=off}

\begin{figure}[!htb]
\centering
\bisubcaptionbox[A]
  {Und eine gaaaanz lange Caption: Teilabbildung A}
  {Subfigure A}[0.4\textwidth]{IMAGE}%
\qquad
\bisubcaptionbox[B]
  {Teilabbildung B}
  {Subfigure B}[0.4\textwidth]{IMAGE}%
\bicaption[Abbildungsverzeichnistitel]
  {Und eine noch viel viel viel
   längere deutsche Beschriftung: Deutscher Titel}
  {Short English heading}
\end{figure}

\end{document}

```

IMAGE	IMAGE
(a) Teilabbildung A	(b) Teilabbildung langer Titel B
<i>(a) Subfigure A</i>	<i>(b) Subfigure long title B</i>

Abbildung 1: Deutscher Titel
Figure 1: English Title

IMAGE	IMAGE
(a) Und eine gaaaanz lange Caption: Teilabbildung A	(b) Teilabbildung B
<i>(a) Subfigure A</i>	<i>(b) Subfigure B</i>

Abbildung 2: Und eine noch viel viel viel längere deutsche Beschriftung: Deutscher Titel
Figure 2: Short English heading

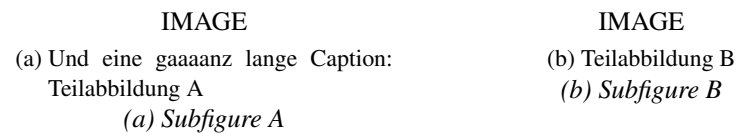


Abbildung 3: Und eine noch viel viel viel längere deutsche Beschriftung: Deutscher Titel
Figure 3: Short English heading

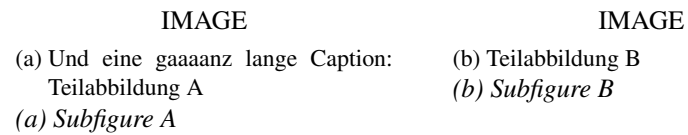


Abbildung 4: Und eine noch viel viel viel längere deutsche Beschriftung: Deutscher Titel
Figure 4: Short English heading